



Research article

 ϵ -Controllability/ ϵ -Observability of linear switched sampled-data systems with the non-equidistant time scheduleAli Hamidoğlu^{a,*}, Gökhan Göksu^b^a Bahçeşehir University, Department of Mathematics, İstanbul, Turkey^b Yıldız Technical University, Mathematical Engineering Department, İstanbul, Turkey

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ABSTRACT

In this work, a novel perspective is developed to investigate the property of controllability/observability of linear switched sampled-data systems under the non-equidistant sampling schedule. In this regard, two new notions are introduced as ϵ -controllability and ϵ -observability to create a feasible ground for controlling and observing linear switched sampled-data systems with any initial time chosen from outside of ϵ neighborhood of switching instants. The main motivation is to control and observe the given switched sampled-data system, which consists of a finite number of discrete-time sub-systems within each sub-system. Hence, the system requires a lower number of sampling times for ϵ -controllability and ϵ -observability compared to the original notions of controllability and observability in this context. Although the number of sampling times in a certain time interval increases whenever ϵ tends to zero, the number of sampling candidates required for ϵ -controllability and ϵ -observability becomes finite. Numerical experiments are performed on several continuous time switched linear systems, and ϵ -controllability and ϵ -observability of their corresponding switched sampled-data systems are derived for various ϵ values under constructed non-equidistant sampling patterns.

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1. Introduction

Switched systems belong to a particular class of hybrid systems which are considered to be continuous time systems coupled with discrete switching events emerging from certain sets. In these systems, the system dynamic is orchestrated with a switching signal at each instant of time among system modes. This phenomenon is mostly seen in real-world problems when obstacles or disturbances occur in their continuous dynamics.

In system analysis and synthesis, justifying whether the system is observable, controllable, or stable occupies a prominent place both in the theory and the application. In this regard, complete geometric conditions for controllability and reachability of switched linear systems are provided and studied in [1–3] and later extended in [4]. Necessary and sufficient conditions are investigated in [5] for observability of linear switched systems with state jumps. An extended version of the controllability problem of switched systems is examined in [6] with the control input constrained in a cone. In addition, both stability analysis and controllability of linear switched systems are presented in [7].

As computers have been involved in the majority of real life processes, designing and investigating digital platforms in

continuous-time system plants have gained much interest, rather than using continuous state feedback tools, which are quite impossible to implement on a computer. Hence, the discretization of a dynamical system by means of time sampling [8] or state sampling [9] becomes a crucial task in application, which allows the machine to evaluate the necessary data within a finite number of periods. In this context, preserving the original properties of continuous plants, such as controllability, observability, or stability after the transfer, has been an interesting field of study. In literature, various observer-based control structures are proposed to preserve the properties of observability [10–12], finite-time stability [13], exponential stability [14], asymptotic stability [15] in sampled-data systems. In this regard, different kinds of sampling patterns are investigated in [16–21] to secure properties of observability, stability, or controllability of sampled-data systems after the process of discretization in an infinite time horizon. Sufficient conditions are derived in [22] for preserving the controllability property of linear switched sampled-data systems under almost any sampling rate. Furthermore, controllability and observability properties of switched linear systems composed of continuous-time and discrete-time sub-systems are studied in [23] under the common sampling period. Moreover, the controllability property of multi-agent systems is examined in [24] with non-pathological sampling time [25]. Recently, an optimal state estimation was designed in [26] for heterogeneous multi-agent systems with networks.

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In this work, problems of controllability and observability of linear switched sampled-data systems are considered by means of the sample and hold method via an non-equidistant time schedule. In our context, we provide the sample and hold technique which deals with the inputs and outputs of the system computed at times $0, \tau, 2\tau, \dots$ (τ as sampling period), and the control stays fixed in $[k\tau, (k+1)\tau)$ for $k \in \mathbb{N}$. This sampling pattern with period τ is built in an equidistant manner [27] and such construction is completely studied in [8,18,28]. During the process of discretization, properties of controllability, observability, or stability of the original system could be lost [8] and the same scenario may occur in the context of switched systems. For instance, we consider the output signal $y(t) = \sin(t\pi\rho(t))$ for the second order observable system with a switching signal $\rho : [0, \infty) \rightarrow \mathcal{V}$, where $\mathcal{V}$ could be a finite or a countable set, and assume that the time is sampled by $t_k = k\tau$ for $k \in \mathbb{N}$. Then, we have $y(t_k) = 0$ for all $k \in \mathbb{N}$, whenever $1/\tau \in \mathcal{V}$ which makes the sampled-data system non-observable. Hopefully, this obstacle could be easily managed by defining a new sampling period τ_0 satisfying $\tau/\tau_0 \in \mathbb{R}/\mathbb{Q}$. Hence, we observe the system with the sampling time $k\tau_0$ whenever the information is not available for $k\tau$ and vice versa. However, this strategy may fail when we cope with a higher order system. As an example, we consider the following signal output

$$y(t) = \sin(t\pi(\tau - \tau_0)\rho(t)) + \sin(t\pi\tau_0\rho(t)) - \sin(t\pi\tau\rho(t)),$$

with the switching signal $\rho : [0, \infty) \rightarrow \mathcal{V}$. Here, we have $y(k\tau) = y(k\tau_0) = 0$ for all $k \in \mathbb{N}$, whenever $2/\tau\tau_0 \in \mathcal{V}$ which makes the system non-observable. Hence, this fact draws our attention to the use of non-equidistant time sampling on switched systems for preserving controllability and observability properties after the transfer. In this context, a simple periodic sampling pattern is designed in [17] with a non-equidistant manner to secure the properties of controllability and observability of continuous-time systems after the discretization procedure. Similar problems are investigated in [29] by means of a non-uniform sampling scheme in sampled-data systems. More recently, a novel non-equidistant sampling schedule has been introduced in [16] which allows one to decrease or increase the maximum allowable sampling period [30] for controllability and observability issues in sampled-data systems. Hence, this strong property highly motivates us to utilize this construction in linear switched sampled-data systems. Here, we see that there is a lack of information on the controllability and observability of the system when the initial sampling data is taken near any switching instant. In this regard, two notions are introduced as ϵ -controllability and ϵ -observability in the context of sampled-data systems, and we show that the discrete case is ϵ -controllable/ ϵ -observable for an arbitrary $\epsilon > 0$ whenever the Kalman rank condition satisfies for the continuous case. Moreover, we realize that the number of sampling times in a certain time gap becomes larger whenever ϵ gets closer to zero. However, this problem could be easily managed by making the upper diameter of sampling times as small as possible. Hence, this framework encourages the application of the ϵ -controllability/ ϵ -observability concept in switched sampled data systems, which allows one to use less number of sampling times when ϵ is properly defined. The main contributions of this work are summarized as follows:

- Notions of ϵ -controllability and ϵ -observability are introduced in the context of switched sampled-data systems.
- ϵ -Controllability/ ϵ -Observability property allows one to control/observe the given switched sampled-data system, which consists of a finite number of discrete-time sub-systems within each sub-system.

- ϵ -Controllability/ ϵ -Observability creates an intuitive point of view to control/observe the given sampled-data system with any initial time taken ϵ apart from the switching instants.
- Kalman rank condition becomes a necessary and sufficient condition to ensure the ϵ -controllability and ϵ -observability of the switched sampled-data system, which provides an easy task to cope with the situation instead of performing sophisticated rank conditions [1] on constant matrices provided in each sub-system.
- ϵ -Controllability/ ϵ -Observability notion coincides with the concept of controllability/observability when ϵ gets closer to zero, and for each ϵ small, it is still possible to construct a non-equidistant sampling schedule for the sampled-data system to secure these notions in discrete time.
- ϵ -Controllability/ ϵ -Observability requires less number of sampling times to control/observe the sampled-data system when it is compared with the original concept of controllability/observability if ϵ is properly defined.
- Different kinds of industrial problems are investigated as continuous time switched linear systems, and for different ϵ values, ϵ -controllability and ϵ -observability properties are achieved for their corresponding switched sampled-data systems under non-equidistant time schedules.

The paper is organized as follows. In Section 2, we provide preliminaries of controllability and observability notions and theorems together with the non-equidistant sampling pattern. In Section 3, we state definitions of ϵ -controllability and ϵ -observability in the context of switched sampled-data systems and prove the main results of the work. In Section 4, we investigate different kinds of controllability and observability problems for continuous time linear switched systems, where their corresponding ϵ -controllability and ϵ -observability properties are derived for various ϵ scenarios in the context of linear switched sampled-data systems. Finally, Section 5 summarizes the paper with further remarks and future work.

2. Preliminaries

In this section, we give the statement of the problem together with definitions and theorems. Moreover, we provide a novel non-equidistant sampling pattern proposed in [16] for preserving the concepts of controllability and observability in linear sampled-data systems.

2.1. Controllability/Observability in finite time

Consider the following continuous linear system in finite time:

$$\Sigma : \begin{cases} \dot{x}(t) = Ax(t) + Bu(t), & t \in (0, T) \\ y(t) = Cx(t), & t \in (0, T) \\ x(0) = x_0, \end{cases} \quad (1)$$

where $T > 0$ is the final time and $A \in \mathbb{R}^{n \times n}$, $B \in \mathbb{R}^{n \times m}$ and $C \in \mathbb{R}^{r \times n}$ are constant matrices. Now, we provide notions of observability and controllability in the bounded domain $(0, T)$ (see, e.g., [31]).

Definition 2.1 (Observability). The system (1) is said to be observable in $(0, T)$ if, for any initial time $t_0 \geq 0$ and state $x(t_0) = x^0$, there exists a finite time $t_f > t_0$ with $t_f < T$ such that the knowledge of $u(t)$ and $y(t)$ for $t_0 \leq t \leq t_f$ suffices to determine x^0 uniquely.

Definition 2.2 (Controllability). The system (1) is said to be controllable in $(0, T)$ if, for any initial time $t_0 \geq 0$ and state $x(t_0) = x^0$ together with the final state x_f , there exists a finite time $t_f > t_0$ with $t_f < T$ and a control $u(t)$ for $t_0 \leq t \leq t_f$ such that $x(t_f) = x_f$.

Here, we recall that the Kalman rank condition is a necessary and sufficient condition for the controllability and observability of the system (1) in finite time T [28].

Lemma 2.3 (Kalman Rank Condition). The system (1) is controllable (or observable) in finite time T if and only if

$$\text{rank}[B, AB, \dots, A^{n-1}B] = n. \quad (2)$$

$$\left(\text{or } \text{rank}[C, CA, \dots, CA^{n-1}]^T = n. \right) \quad (3)$$

2.2. A time sampling schedule

In this part, we provide the sampling pattern constructed in the recent work [16] which consists of two parts. First of all, we choose a positive irrational number $\xi \in \mathbb{R}^+$ and an integer $\eta \in \mathbb{N}$ with $\eta \geq n$, satisfying

$$\eta\xi < 1 < (\eta + 1)\xi, \quad (4)$$

where n stands for the dimension of state space of Σ . In the first part, we choose sampling times $t_0, t_1, \dots, t_{\rho_n+n}$, satisfying the following relation:

$$t_\mu = (k - (\mu - \rho_k))\xi + \frac{(\mu - \rho_k)}{\eta} \quad \text{where } \rho_k \leq \mu < \rho_{k+1},$$

and ρ_k represents the sum of the first k positive integers, with $\rho_0 = 0$. Hence, we have totally $\frac{n(n+3)}{2}$ number of sampling times in the first part.

In the second part, we construct the rest of the pattern $t_{\rho_n+1}, t_{\rho_n+1+1}, \dots, t_{n^2+1}$ by adopting the following strategy

$$t_j^i = (n-j)\xi + (i+j)\frac{1}{\eta} \quad \text{for } 2 \leq i+j \leq n-1,$$

and i, j positive integers with the adjustment

$$t_j^i = t_{\rho_{n+1}-\rho_{i-1}+(i-1)(n-1)+j-1}, \quad (5)$$

which would be a continuation of the time sequence constructed in the first part. Hence, we have $t_0, t_1, \dots, t_{\rho_n+n}$ and $t_{\rho_n+1}, t_{\rho_n+1+1}, \dots, t_{n^2+1}$ sampling times from the first and second parts respectively, all together $n^2 + 2$ sampling times at the end. Denote the set of those points as

$$P := \{t_0, t_1, \dots, t_{n^2+1}\}. \quad (6)$$

For reaching the whole positive real line, the set P periodically extended as follows:

$$t_\mu = mt_{n^2+1} + t_k \quad \text{where } \mu = m(n^2 + 1) + k. \quad (7)$$

Define the following collection of sets

$$P_m := \{t_{m(n^2+1)}, t_{m(n^2+1)+1}, \dots, t_{(m+1)(n^2+1)}\}, \quad (8)$$

for $m \in \mathbb{N}$. Observe that, for $m = 0$, we have $P_0 = P$ as defined in (6). Moreover, we represent the whole sampling as the union of those sets, i.e.,

$$\bigcup_{i=0}^{\infty} P_i = \{t_i\}_{i=0}^{\infty}, \quad (9)$$

by definition. In other words, we have

$$P_m = \{t_{m(n^2+1)} + t_i : t_i \in P\}. \quad (10)$$

Theorem 2.4. For any ϵ small and positive real number α , the open interval $(\alpha, \alpha + \epsilon)$ contains at least one set of the form (8) if

$$\eta > \frac{4(n-1)}{\epsilon}. \quad (11)$$

Proof. Take any ϵ and α and fix them. Here, the sampling schedule is constructed as the union of sets P_m , each of which shares the common gap

$$t_{(m+1)(n^2+1)} - t_{m(n^2+1)} = t_{n^2+1}, \quad (12)$$

for each $m \in \mathbb{N}$. Then, under the condition of (11), it suffices to prove that

$$t_{n^2+1} < \frac{\epsilon}{2}. \quad (13)$$

From the condition (4), we obtain

$$t_{n^2+1} \leq \frac{2(n-1)}{\eta} < \frac{\epsilon}{2}, \quad (14)$$

which concludes the proof. $\square$

Lastly, we give the following three lemmas proposed in [16]. The first lemma is known as the extended case of Kronecker's density theorem in $\mathbb{R}$ [16].

Lemma 2.5. Let α and β be positive real numbers such that $\frac{\alpha}{\beta}$ is an irrational number. Then, the set

$$K := \{j\beta + i\alpha \mid i, j \in \mathbb{Z}\}, \quad \text{is dense in } \mathbb{R}.$$

Now, we provide the second lemma, which is the extended version of the Cayley–Hamilton theorem [16]:

Lemma 2.6. Let A be an $n \times n$ invertible matrix, then for any $i, j \in \mathbb{Z}$, one can have

$$A^j = \alpha_0 A^i + \alpha_1 A^{i+1} + \dots + \alpha_{n-1} A^{i+n-1}$$

where α_i some real numbers for $i = 0, 1, \dots, n-1$.

The following last lemma is related to protecting controllability and observability properties under the non-equidistant sampling schedule (9) in linear sampled-data systems [16].

Lemma 2.7. Properties of controllability and observability of the continuous-time system (1) are preserved under the sample and hold operation by means of the sampling pattern (9) in an infinite time horizon.

2.3. Linear switched sampled-data systems

Consider the following continuous linear switched system:

$$\Sigma : \begin{cases} \dot{x}(t) = A_{\sigma(t)}x(t) + B_{\sigma(t)}u(t), & t > 0 \\ y(t) = C_{\sigma(t)}x(t), & t > 0 \\ x(0) = x_0, \end{cases} \quad (15)$$

where $\sigma : [0, \infty) \rightarrow \mathcal{V}$ represents the switching signal of the system, which is a piecewise constant function with a finite index set $\mathcal{V}$, $A_{\sigma(t)} \in \mathbb{R}^{n \times n}$, $B_{\sigma(t)} \in \mathbb{R}^{n \times m}$ and $C_{\sigma(t)} \in \mathbb{R}^{r \times n}$ are constant matrices depends on the switching signal. Moreover, $x(t)$ and $u(t)$ represent the state and control of the system respectively. Here, we assume that there are $k > 0$ total switching instants acting on the system. Let $s_0 := 0$, and define the set of switching instants as

$$S := \{s_i\}_{i \in I_k}, \quad \text{where } I_k := \{1, \dots, k\}. \quad (16)$$

Moreover, we define the minimum distance between consecutive switching instants as

$$s_D = \min_{i \in I_k} (s_i - s_{i-1}), \quad (17)$$

which is also known as “dwell-time” in switched systems (see, e.g., [32]). Without loss of generality, we define

$$A_i := A_{\sigma(t)}; \quad B_i := B_{\sigma(t)}; \quad C_i := C_{\sigma(t)}, \quad (18)$$

where $t \in [s_{i-1}, s_i]$ with $i \in I_k$. Now, we consider the following continuous sub-systems defined for each switching instant:

$$\Sigma_i : \begin{cases} \dot{x}(t) = A_i x(t) + B_i u(t), & t \in [s_{i-1}, s_i] \\ y(t) = C_i x(t), & t \in [s_{i-1}, s_i] \\ x(s_{i-1}) = x_{s_{i-1}}, \end{cases} \quad (19)$$

for $i \in I_k$. Observe that the corresponding solution of each system Σ_i would be of the form

$$x(t) = e^{A_i(t-s_{i-1})} x_{s_{i-1}} + \int_{s_{i-1}}^t e^{A_i(t-\tau)} B_i u(\tau) d\tau. \quad (20)$$

Discretization of the system Σ under zero-order hold operation with the sampling $\{t_i\}_{i=0}^\infty$ would be

$$\Sigma^d : \begin{cases} x(t_{j+1}) = A_{\sigma(t_j)} x(t_j) + B_{\sigma(t_j)} u_j \\ y(t_j) = C_{\sigma(t_j)} x(t_j), \end{cases}$$

where u_j is a control input which is constant along the interval $[t_j, t_{j+1})$. Hence, each sub-system of the sampled-data system Σ^d becomes

$$\Sigma_i^d : \begin{cases} x_{k+1} = e^{A_i(t_{k+1}-t_k)} x_k + \int_{t_k}^{t_{k+1}} e^{A_i(t_{k+1}-\tau)} B_i u_k d\tau \\ y_k = C_i x_k, \end{cases}$$

where $x_k := x(t_k)$, $y_k := y(t_k)$, and u_k , which is the control input which is constant along the interval $[t_k, t_{k+1})$.

2.4. Equidistant versus non-equidistant: A motivating example

In this part, we consider the following controllable three dimensional linear system [33]:

$$\dot{x}(t) = Ax(t) + Bu(t), \quad (21)$$

on the interval $(0, \pi\omega)$ where $\omega > 1$ is a positive real number and the matrices A and B are defined as

$$A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 5 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \end{bmatrix}.$$

The corresponding sampled-data system under an equidistant sampling period $T = \pi$ is given by

$$x_{k+1} = A_T x_k + B_T u_k \quad (22)$$

where

$$x_k = x(k\pi), \quad u_k = u(k\pi), \quad A_T = e^{A\pi}, \quad B_T = \int_0^\pi e^{\tau A} d\tau B.$$

For this equidistant sampling schedule, It can be verified that

$$\text{rank}\{B_T, A_T B_T, A_T^2 B_T\} = 2 < 3$$

which is less than the full rank. In other words, for a given $\omega > 1$, sampled-data system (22) becomes uncontrollable on $(0, \pi\omega)$. In this example, we see that the controllability property of sampled-data system is not achieved by means of periodic time sampling with period $T = \pi$. This observation motivates us to use a non-equidistant time schedule for the sampled-data system without

losing controllability/observability property of the original system, namely, the Kalman rank condition (2)/(3) satisfying for the sampled data system. Building such a time schedule, we try not to choose the initial time for the controllability and observability problem from the interval with a small gap length, whose points are close to the end point $\pi\omega$. In the following section, we state two novel notions of controllability and observability in switched sampled-data systems from this perspective and prove that these properties are preserved under the non-equidistant sampling pattern whose construction depends on the given gap length.

3. ϵ -Controllability/ ϵ -Observability

Let $\epsilon > 0$ be given such that $\epsilon < s_D$. Throughout this paper, we fix ϵ and concentrate on the number η satisfying the following inequality:

$$\eta > \frac{4(n-1)}{\epsilon}. \quad (23)$$

According to Theorem 2.4, for any positive real number α , the open interval $(\alpha, \alpha + \epsilon)$ contains at least one set of the form (8). Moreover, the inequality (23) is valid for any $\epsilon > 0$, since η could be taken as a large number satisfying (4) for some irrational ξ .

Given time schedule $\{t_i\}_{i=0}^\infty$, construct the following sets:

$$\begin{aligned} J_{s_i} &:= [s_{i-1}, s_i) \cap \{t_i : i \in \mathbb{N}\}, \\ J_{s_i}^\epsilon &:= [s_{i-1}, s_i - \epsilon) \cap \{t_i : i \in \mathbb{N}\}, \\ J^\epsilon &= \bigcup_{i \in I_k} J_{s_i}^\epsilon \quad \text{and} \quad J_s = \bigcup_{i \in I_k} J_{s_i}. \end{aligned} \quad (24)$$

Observe that $J^\epsilon \subset J_s$ and $J^\epsilon = J_s$ whenever $\epsilon = 0$. Now, we are ready to provide the notions of ϵ -observability and ϵ -controllability of the switched sampled-data system Σ^d .

Definition 3.1 (ϵ -Observability). The system Σ^d is said to be ϵ -observable if for any initial time $t_q \in J^\epsilon$, there exists a positive integer ν (number of steps), with $t_{q+\nu} \in J_s$ such that the state x_q is uniquely determined from $\{y_q, y_{q+1}, \dots, y_{q+\nu}\}$ and $\{u_q, u_{q+1}, \dots, u_{q+\nu-1}\}$.

Definition 3.2 (ϵ -Controllability). The system Σ^d is said to be ϵ -controllable, if for every initial time $t_k \in J^\epsilon$ and given any final states x^1 , there exists a positive integer ν (number of steps) with $t_{k+\nu} \in J_s$ and a control sequence $\{u_k, u_{k+1}, \dots, u_{k+\nu-1}\}$ which derives $x_{k+\nu}$ to the final state x^1 , namely, $x_{k+\nu} = x^1$.

In this context, we focus on the controllable/observable continuous-time sub-system (19) for each $i \in I_k$. Therefore, we assume that the matrices A_i, B_i and A_i, C_i satisfy the Kalman rank condition for each $i \in I_k$ which makes the continuous-time switched system Σ to become controllable/observable in finite time. Now, we state the following main results, which simply point out that the property of observability and controllability of the continuous time switched system Σ can be carried over to a switched sampled-data system Σ^d by means of the sampling schedule.

Theorem 3.3 (ϵ -Observability). Assume that the matrices A_i and C_i satisfy the Kalman rank condition for each $i \in I_k$. Then, the switched sampled-data system Σ^d is ϵ -observable under the sampling schedule (9).

Proof. Here, we prove the statement by using proof by contradiction, i.e., assuming that the discrete system Σ^d is not ϵ -observable. Hence, contrary to what the statement would turn out to be, for some initial time $t_p \in J^\epsilon$, the state x_p cannot be

uniquely determined from the set J_s . More precisely, let $t_p \in J_{s_j}^\epsilon$ for some $j \in I_k$, then each of the following equations

$$y_m = C_j e^{A_j(t_m - t_p)} x_p, \quad \text{for any } t_m \in J_{s_j} \text{ with } m \geq p,$$

has no unique solution $x(t_p) = x_p \in \mathbb{R}^n$. This means that, there exists a non-zero vector $\theta \in \mathbb{R}^n$ such that

$$C_j e^{A_j(t_m - t_p)} \theta = 0 \quad \text{for } t_m \in J_{s_j} \text{ with } m \geq p. \quad (25)$$

By applying [Theorem 2.4](#) on the interval $(s_j - \epsilon, s_j)$ which contains time samplings from the set $J_{s_j} \setminus J_{s_j}^\epsilon$, one can get that $P_\ell \subset J_{s_j} \setminus J_{s_j}^\epsilon$ for some $\ell \in \mathbb{N}$. In other words,

$$C_j e^{A_j t_m} \phi = 0 \quad \text{for } t_m \in P, \quad (26)$$

where

$$\phi := e^{A_j(t_{\ell(n^2+1)} - t_p)} \theta.$$

Note that $t_{\rho_{i+1}-1} = \frac{i}{\eta} \in P$ for $i = 0, 1, \dots, n-1$ and the matrix $e^{A_j \frac{1}{\eta}}$ is non-singular then, from [Lemma 2.6](#), for any $v \in \mathbb{Z}$, we obtain the following:

$$C_j e^{A_j \frac{v}{\eta}} \phi = C_j \sum_{k=0}^{n-1} \alpha_k e^{A_j \frac{k}{\eta}} \phi = 0, \quad (27)$$

for some $\alpha_k \in \mathbb{R}$ with $k \in \{0, 1, \dots, n-1\}$. Moreover, we observe that $t_{\rho_i} = i\xi \in P$ for $i = 0, 1, \dots, n-1$. Similarly, by using [Lemma 2.6](#), for any $v \in \mathbb{Z}$, we have the following

$$C_j e^{A_j v\xi} \psi = 0. \quad (28)$$

By using [Lemma 2.6](#) and (27) together with (28), for every $v, w \in \mathbb{Z}$, we obtain

$$C_j e^{A_j(\frac{v}{\eta} + w\xi)} \phi = \sum_{r,k=0}^{n-1} \alpha_r \beta_k C_j e^{A_j(\frac{r}{\eta} + k\xi)} \phi = 0, \quad (29)$$

for some $\alpha_r, \beta_k \in \mathbb{R}$ with $r, k \in \{0, 1, \dots, n-1\}$. Here, we should note that

$$\frac{r}{\eta} + k\xi \in P \quad \text{for } r, k \in \{0, 1, \dots, n-1\}.$$

According to [Lemma 2.5](#), the set $\{\frac{v}{\eta} + w\xi : v, w \in \mathbb{Z}\}$ becomes dense in $\mathbb{R}$ which concludes from (29) and analyticity of exponential function that

$$C_j e^{A_j t} \phi = 0 \quad \text{for all } t \geq 0, \quad (30)$$

which contradicts the fact that Σ_j is observable. Hence, the discrete system Σ^d is ϵ -observable. $\square$

Using the relationship between the Kalman rank criteria for controllability and observability, we now propose the following conclusion for ϵ -controllability, which is a direct consequence of the prior theorem for ϵ -observability. In this theorem, it is assumed that the matrices A_j and B_j meet the Kalman rank condition for each $j \in I_k$. Under the stated sample schedule, the theorem says that the system is ϵ -controllable, i.e., it can be controlled within a specific tolerance level. This result has major implications for the design of control systems for switched sampled-data systems and provides a valuable instrument for measuring the controllability of such systems.

Theorem 3.4 (ϵ -Controllability). Assume that the matrices A_j and B_j satisfy the Kalman rank condition for each $j \in I_k$. Then, the switched sampled-data system Σ^d is ϵ -controllable under the sampling schedule (9).

Proof. Here, we make use of the duality concept between observability and controllability in this proof. Observe that for any $j \in I_k$, we have

$$\text{rank}[B_j, \dots, A_j^{n-1} B_j] = \text{rank}[B_j^\top, \dots, B_j^\top (A_j^\top)^{n-1}]^\top. \quad (31)$$

Let $C_j = B_j^\top$ and $\tilde{A}_j = A_j^\top$. Substituting these matrices in (31), we have

$$\text{rank}[B_j, \dots, A_j^{n-1} B_j] = \text{rank}[C_j, \dots, C_j \tilde{A}_j^{n-1}]^\top.$$

Hence, the Kalman rank condition for A_j and B_j implies ϵ -observability of the system Σ^d with matrices C_j and $\tilde{A}_j$ for all $j \in I_k$, according to [Theorem 3.3](#). But, this means that any $\phi \in \mathbb{R}^n$ satisfying

$$C_j e^{\tilde{A}_j t_m} \phi = 0 \quad \text{for } t_m \in P \text{ and } j \in I_k, \quad (32)$$

implies that $\phi = 0$. By substituting $B_j^\top$ and $A_j^\top$ in (32) and taking transpose from both sides, we obtain

$$\phi^\top e^{A_j t_m} B_j = 0 \quad \text{for } t_m \in P \text{ and } j \in I_k, \quad (33)$$

implies that $\phi = 0$ according to ϵ -observability of the system. Since, this duality is valid for all $j \in I_k$, the system Σ^d becomes ϵ -controllable. $\square$

In a case of infinite time horizon, ϵ -controllability/ ϵ -observability follows directly from [Lemma 2.7](#):

Corollary 3.5. If the continuous-time system Σ is observable/controllable in $[s_k, \infty)$ then, its sampled-data system is observable/controllable in $[s_k, \infty)$ with the sampling schedule $\{t_i\}_{i=0}^\infty$.

The following result points out that the number of steps required for ϵ -controllability/ ϵ -observability becomes finite. This fact comes from [Theorem 2.4](#) that, for any given ϵ , one can build a time sampling schedule $\{t_i\}_{i=0}^\infty$ in such a way that any interval with length ϵ would contain a complete set of the form (8). Hence, for some $m \in \mathbb{N}$, having the whole set P_m as its input guarantees the system to become ϵ -controllable/ ϵ -observable.

Corollary 3.6. Let ν be the minimum number of steps for securing the controllability/observability of a sampled-data system, then the following

$$n^2 + 1 \leq \nu \leq 2(n^2 + 1), \quad (34)$$

holds

Remark 1. In Section 2.4, for any $\epsilon > 0$ and $\omega > 1$ given, the sampled-data system becomes ϵ -controllable under the non-equidistant sampling pattern (9). This comes from the fact that the original system satisfies the Kalman rank condition, which makes the sampled-data system ϵ -controllable by [Theorem 3.4](#). For example, take $\epsilon = 0.1$ and $\omega = 5$. Then, the sampled-data system is ϵ -controllable with the following sampling schedule:

$$P_m = \{t_{10m} + t_i : t_i \in P\} \text{ with } \eta = 100, \xi = \frac{\sqrt{2}}{142}, \quad (35)$$

where the set P is described in [Fig. 1](#) for $n = 3$. Under this sampling construction, we obtain

$$P_{391} \subset [5\pi - 0.1, 5\pi), \quad (36)$$

which establishes the ϵ -controllability of the system when $\epsilon = 0.1$.

Remark 2. For any $\epsilon > 0$ and time instant T , one can build various non-equidistant sampling patterns (9) with main candidates η and ξ satisfying both (4) and (23) such that

$$P_K \subset [T - \epsilon, T), \quad (37)$$

holds for some $K \in \mathbb{N}$.

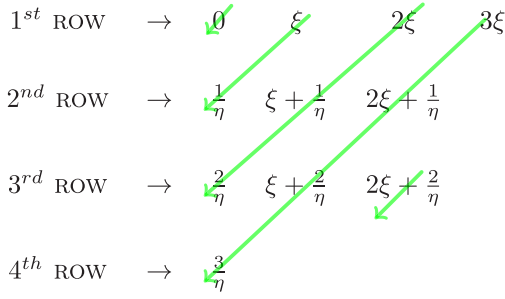


Fig. 1. 3-D Construction of the sampling pattern P arranged as ascending order ($n = 3$).

Remark 3. In Corollary 3.5, we see that the transfer from continuous to discrete case is exact, i.e., $\epsilon = 0$. Hence, the concept of 0-observability/0-controllability coincides with the original notion of observability/controllability in sampled-data systems. Moreover, we see that the property of ϵ -observability/ ϵ -controllability is slightly weaker than the original one. Hence, observability/controllability implies ϵ -observability/ ϵ -controllability in the context of the sampled-data system.

Remark 4. In both Theorems 3.3 and 3.4, we see that the discrete system Σ_i^d has an absence of controllability/observability issues near the boundary s_i for each $i \in I_k$. Here, $\epsilon > 0$ could be taken so small that one can possibly get controllability or observability for the whole interval. However, when ϵ tends to zero, the question arises whether it is still possible to find a set P_ℓ such that

$$P_\ell \subset [s_i - \epsilon, s_i],$$

for some $\ell \in \mathbb{N}$. Hopefully, the answer is affirmative for this query thanks to the sampling pattern, which makes it feasible to set the upper diameter as small as possible.

Remark 5. As it is discussed in Remark 3, ϵ -controllability/ ϵ -observability notion coincides with the concept of controllability/observability when ϵ approaches to zero. In this regard, one could deal with a huge amount of sampling times in a certain time interval, which would definitely take time and effort to implement on a digital platform. This fact is the motivating reason that promotes the use of ϵ -controllability and ϵ -observability concepts in sampled-data systems, which allows one to use fewer sampling times when ϵ is properly defined according to the choice of initial time.

4. Illustrative examples

In this part, we investigate ϵ -controllability and ϵ -observability of some linear switched sampled-data systems under the non-equidistant sampling schedule (9) for different values of ϵ by using numerical simulations.

4.1. A switching 2-D pendulum

Consider the following second order control problem:

$$\ddot{\phi}(t) + \delta(t)\phi(t) = u(t), \quad (38)$$

where $\delta(t)$ is a switching signal, $\phi(t)$ is the angle of displacement of the pendulum, and $u(t)$ is the control. The original controllability problem is to find a proper control for deriving $(\phi, \dot{\phi})$ to desired state $(0, 0)$ for all small initial data without interrupting its nature. Here, we emphasize on this problem where the pendulum is hung to a swinging support instead of a persistent wall.

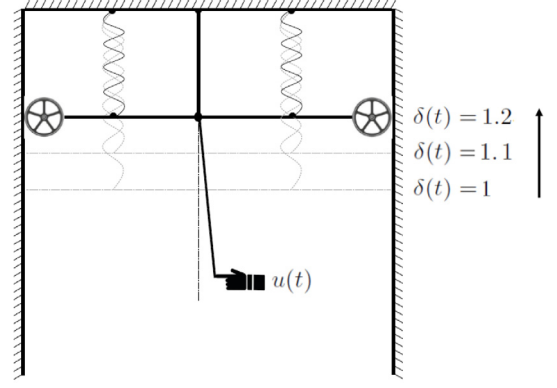


Fig. 2. A pendulum with swinging support.

In this regard, the strength of the force applied by the swinging support is changing in a switching manner (see, e.g., Fig. 2). A finite dimensional form of (38) would be

$$\dot{x}(t) = A_{\sigma(t)}x(t) + Bu(t), \quad t \geq 0, \quad (39)$$

where

$$A_{\sigma(t)} = \begin{bmatrix} 0 & 1 \\ -\delta(t) & 0 \end{bmatrix}, \quad B = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \quad (40)$$

with the state $x(t) = [\phi(t) \quad \dot{\phi}(t)]^T \in \mathbb{R}^2$ for $t \geq 0$. Observe that Kalman's rank condition satisfies for any values of $\delta(t)$. Hence, the continuous time system (39) becomes controllable for any final time.

In this part, we design our sampling schedule $\{t_i\}_{i=0}^\infty$ in such a way that the discrete version of the system under zero-order hold, satisfies the ϵ -controllability property for different scenarios of ϵ values. Let the set of switching instants be given as $S := \{5, 10, 15\}$ and

$$\delta(t) = \left\{ 1 + \frac{i}{10} : t \in [5i, 5(i+1)), i = 0, 1, 2 \right\}, \quad (41)$$

with

$$\delta(t) = 1.25 \quad \text{for } t \in [15, 20]. \quad (42)$$

As it is seen in Fig. 2, the switching signals are emerged by means of swinging support. Note that (39) is a two-dimensional switched system, which implies $n = 2$.

4.1.1. First scenario: $\epsilon = 1$

Here, we construct the non-equidistant sampling schedule (9) for $n = 2$ and $\epsilon = 1$. Consider the set

$$P = \left\{ 0, \xi, \frac{1}{\eta}, 2\xi, \xi + \frac{1}{\eta}, \frac{2}{\eta} \right\}. \quad (43)$$

By simply choosing

$$\eta = 5 \quad \text{and} \quad \xi = \frac{e}{14}, \quad (44)$$

where e is an Euler constant and both satisfying inequalities (4) and (23), we obtain

$$P_{10} \subset [4, 5); \quad P_{23} \subset [9, 10); \quad P_{35} \subset [14, 15),$$

as desired. According to Theorem 3.4, the corresponding sampled-data system becomes 1-controllable under the sampling $\{t_i\}_{i=0}^\infty$ satisfying (44).

Now, let us simply perform the following periodic switching controller

$$u(t) = \begin{cases} -1/2, & t \in [0, 5) \cup [10, 15) \cup [20, \infty) \\ 1/2, & t \in [5, 10) \cup [15, 20), \end{cases} \quad (45)$$

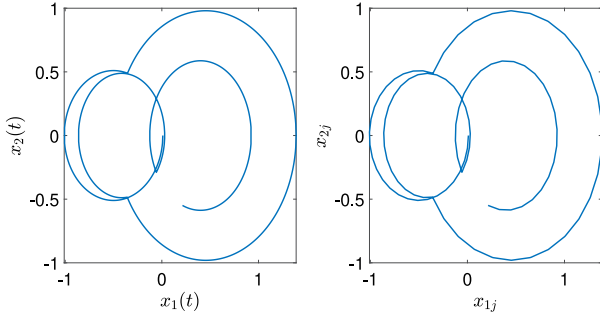


Fig. 3. Phase Portrait of the System (39) with Control Input (45): (Left) Continuous-Time System (Right) Discretization under Sampling $\{t_i\}_{i=0}^{\infty}$ with (44).

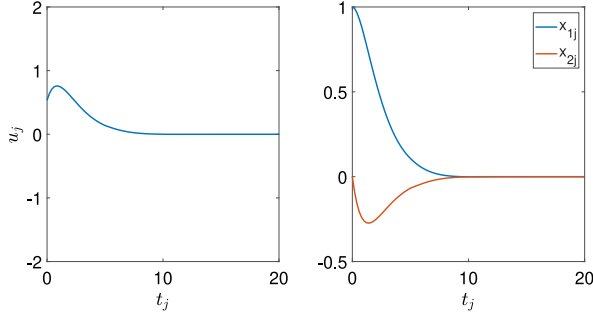


Fig. 4. Stabilizing State Feedback and State Space Response of the System (39) under Sampling $\{t_i\}_{i=0}^{\infty}$ with (44).

for the system (39) which carries the initial state $x(0) = [0.01 \ 0]^T$ to $x(20) = [0.2135 \ -0.5492]^T$. By using the sampling pattern $\{t_i\}_{i=0}^{\infty}$ with main candidates satisfying (44), switching discrete controls $u_j := u(t_j)$ derive the initial state $x_0 = [0.01 \ 0]^T$ to the final state $x^1 = x_{251} = [0.2133 \ -0.5485]^T$ which can be seen in Fig. 3.

For further analysis, we design the state feedback controller $u(t) = Kx(t)$ that stabilizes the continuous time system (39). A sufficient condition (see, e.g., [34]), for the existence of such a controller is guaranteed by finding a symmetric positive definite matrix $P \in \mathbb{R}^{2 \times 2}$ and a matrix $Y \in \mathbb{R}^{1 \times 2}$ such that

$$A_p P + P A_p^T + B Y + Y^T B^T \leq 0, \quad (46)$$

is satisfied for $p = 1, 2, 3, 4$. Note that $\leq$ denotes negative definiteness for matrices. The desired feedback gain can be calculated as $K = Y P^{-1}$ and can be numerically estimated via the MATLAB LMI Toolbox as

$$K = [0.5345 \quad -1.1931]. \quad (47)$$

Numerical simulations show that the discrete state feedback $u_j = Kx_j$ stabilizes the switched sampled data system as shown in Fig. 4 for an initial state $x_1 = [1 \ 0]^T$.

4.1.2. Second scenario: $\epsilon = 0.1$

By choosing the main candidates η and ξ as

$$\eta = 40 \quad \text{and} \quad \xi = \frac{\sqrt{3}}{70}, \quad (48)$$

we obtain that

$$P_{98} \subset [4.9, 5); \quad P_{198} \subset [9.9, 10); \quad P_{298} \subset [14.9, 15],$$

as expected. Therefore, we have the 0.1-controllability for the sampled-data system under the sampling schedule with candidates satisfying (48).

Table 1
An aircraft's parameter values.

p	1	2	3
α_1	-2.93	-2.637	-3.22
α_2	-4.75	-4.275	-5.225
α_3	0.78	0.71	0.85
α_4	0.086	0.08	0.09
α_5	-0.11	-0.099	-0.121
α_6	0.1	0.09	0.11
α_7	-0.042	-0.038	-0.046
α_8	2.601	2.35	2.8
α_9	-0.26	-0.234	-0.28
α_{10}	-3.91	-3.52	-4.3
α_{11}	0.035	0.032	0.038
α_{12}	-2.5335	-2.28	-2.78
α_{13}	0.31	0.28	0.34

4.1.3. Third scenario: $\epsilon = 0.01$

In this case, by selecting the main candidates η and ξ as

$$\eta = 400 \quad \text{and} \quad \xi = \frac{\sqrt{2}}{566}, \quad (49)$$

satisfying (4) and (23), we have

$$P_{998} \subset [4.99, 5); \quad P_{1998} \subset [9.99, 10); \quad P_{2998} \subset [14.99, 15],$$

which provides the 0.01-controllability for the sampled-data system.

4.2. 4-D switching motion model of an aircraft

In this part, we consider the following model, which describes the motion of an aircraft in 4-D:

$$\dot{x}(t) = A_{\sigma(t)} x(t) + B_{\sigma(t)} u(t), \quad t \geq 0, \quad (50)$$

where $\sigma(t)$, $x(t) = [x_1(t) \ x_2(t) \ x_3(t) \ x_4(t)]^T$ and $u(t) = [u_1(t) \ u_2(t)]^T$ represent the switching signal, the state, and the control of an aircraft at time t respectively. More precisely, each component is defined as follows: x_1 is the bank angle; x_2 is its derivative; x_3 is the sideslip angle; x_4 is the yaw rate; u_1 is the rudder deflection; and u_2 is the aileron deflection. In this context, the original model was studied in [35] as a linear parameter varying system, where the nominal value and the boundary values of the system parameters are considered.

In this problem, we consider the family of differential equations

$$\dot{x}(t) = A_p x(t) + B_p u(t), \quad t \geq 0, \quad (51)$$

for $p = 1, 2, 3$ and the matrices A_p and B_p are defined as

$$A_p = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & \alpha_{1,p} & \alpha_{2,p} & \alpha_{3,p} \\ \alpha_{4,p} & 0 & \alpha_{5,p} & -1 \\ \alpha_{6,p}\alpha_{4,p} & \alpha_{7,p} & \alpha_{8,p} + \alpha_{5,p}\alpha_{6,p} & \alpha_{9,p} - \alpha_{6,p} \end{bmatrix},$$

and

$$B_p = \begin{bmatrix} 0 & 0 \\ 0 & \alpha_{10,p} \\ \alpha_{11,p} & 0 \\ \alpha_{12,p} + \alpha_{6,p}\alpha_{11,p} & \alpha_{13,p} \end{bmatrix}.$$

The parameter values of the aircraft is given in Table 1.

In our case, we consider four switching procedures as shown in Fig. 5.

More precisely,

$$\sigma(t) = \begin{cases} 1, & t \in [0, 2) \cup [2.1, 4) \cup [4.2, 5] \\ 2, & t \in [2, 2.1) \\ 3, & t \in [4, 4.2) \end{cases} \quad (52)$$

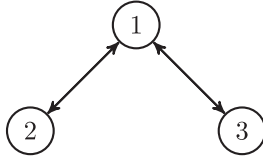
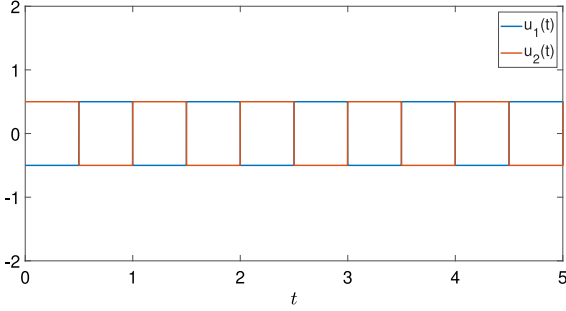
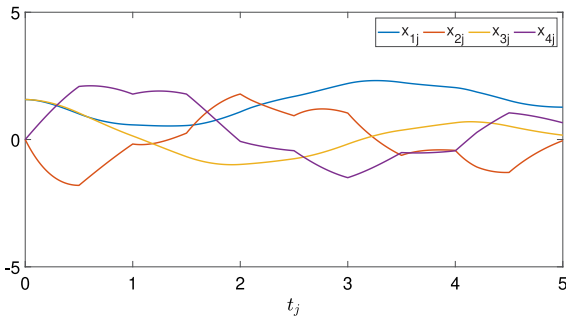


Fig. 5. Switching Procedures.

Fig. 6. Periodical Switching Controller $u(t)$.Fig. 7. State Space Response of the System (50) with Control Input depicted in Fig. 6 under Sampling $\{t_i\}_{i=0}^{\infty}$ with (54).

For each family of differential Eqs. (51), we see that A_p and B_p satisfy the Kalman rank condition. In this case, we focus on the ϵ -controllability problem when $\epsilon = 0.01$. In this regard, we use the 4-D construction of the sampling pattern:

$$P = \left\{ i\xi + \frac{j}{\eta} : i, j \in \{0, 1, 2, 3\} \right\} \cup \left\{ 4\xi, \frac{4}{\eta} \right\}, \quad (53)$$

which has eighteen elements in total. From Theorem 2.4 and (4), we set

$$\eta = 1250 \quad \text{and} \quad \xi = \frac{e}{3400}, \quad (54)$$

Then, we have

$$P_{415} \subset [1.99, 2); P_{436} \subset [2.09, 2.1); P_{832} \subset [3.99, 4); \\ P_{874} \subset [4.19, 4.2); P_{1040} \subset [4.99, 5].$$

Hence, the switched sampled-data system becomes the 0.01-controllable with the sampling pattern (9) associated with (54). Again, we consider the periodical switching controller $u(t)$ for the continuous time system (50) shown in Fig. 6.

By using the constructed sampling schedule $\{t_i\}_{i=0}^{\infty}$, we obtain discrete switching controls $u_j := u(t_j)$ which derives $x_1 = [\pi/4 \ 0 \ \pi/4 \ 0]^T$ to the final state $x_{17715} = [1.2696 \ -0.0435 \ 0.1711 \ 0.6588]^T$ which is numerically demonstrated in Fig. 7.

Now, take the state feedback controller $u(t) = Kx(t)$ in such a way that the continuous time system (50) becomes stable. By

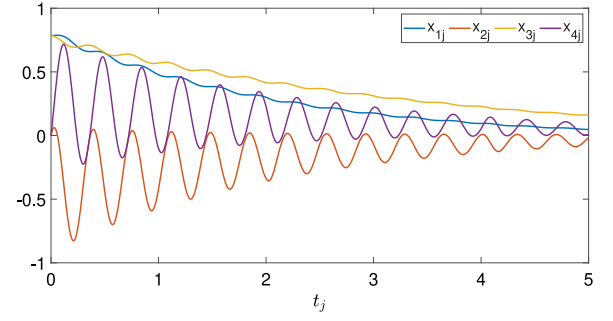
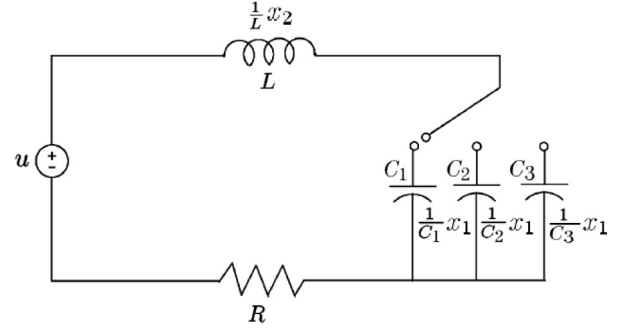
Fig. 8. Stabilizing State Feedback and State Space Response of the System (50) under Sampling $\{t_i\}_{i=0}^{\infty}$ with (54).

Fig. 9. A Switched RLC Circuit.

using a similar analysis, one can obtain a symmetric positive definite matrix $P \in \mathbb{R}^{4 \times 4}$ and a matrix $Y \in \mathbb{R}^{2 \times 4}$ satisfying

$$A_p P + P A_p^T + B_p Y + Y^T B_p^T \leq 0, \quad (55)$$

for $p = 1, 2, 3$. The feedback gain can be calculated numerically via the MATLAB LMI Toolbox as

$$K = \begin{bmatrix} -3.4485 & -8.0455 & -0.2776 & 1.0903 \\ 0.0777 & -0.6476 & -2.7690 & 4.8456 \end{bmatrix}. \quad (56)$$

Numerical simulations show that the discrete state feedback $u_j = Kx_j$ also stabilizes the sampled-data system for an initial state $x_1 = [\pi/4 \ 0 \ \pi/4 \ 0]^T$ as seen in Fig. 8.

4.3. Switched RLC circuit

In this part, we consider a switched RLC circuit which consists of an input power source, a resistance, an inductance, and three capacitors, as presented in Fig. 9.

In this regard, we introduce a periodic switching schedule as a result of a switching among three capacitors with a period $\tau = 20$ s. To this aim, the system matrices of the switched system (15) would be

$$A_{\sigma(t)} = \begin{bmatrix} 0 & \frac{1}{L} \\ -\frac{1}{C(t)} & -\frac{R}{L} \end{bmatrix}, \quad B_{\sigma(t)} = B = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \\ C_{\sigma(t)} = C = \begin{bmatrix} 0 & \frac{1}{L} \end{bmatrix}, \quad (57) \\ c(t) = \begin{cases} c_1, & t \in [0, 20) \cup [60, 80) \cup \dots \\ c_2, & t \in [20, 40) \cup [80, 100) \cup \dots \\ c_3, & t \in [40, 60) \cup [100, 120) \cup \dots \end{cases}$$

Here, the state variables are the charge in the capacitor x_1 and the flux in the inductance x_2 , on the other hand, the input u is the voltage and c_i is the i th capacitor.

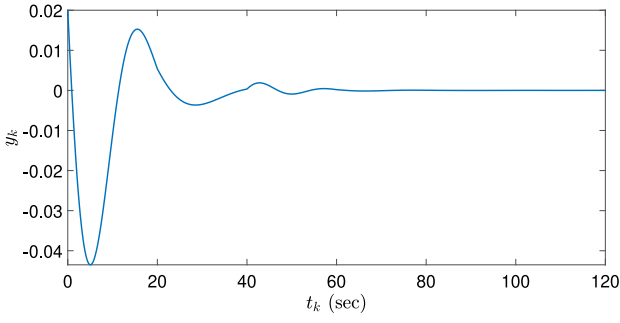


Fig. 10. The Measured Output y_k .

To perform a simulation example, we set $L = 10$ H, $c_1 = 1$ F, $c_2 = 2$ F, $c_3 = 0.5$ F and $R = 1$ Ω . As a control input, we employ an output feedback of the form $u = -y$. Hence, we have the following closed-loop system:

$$\begin{aligned} \dot{x}(t) &= \bar{A}_{\sigma(t)} x(t), \\ y(t) &= Cx(t) \end{aligned} \quad (58)$$

where $\bar{A}_{\sigma(t)} = A_{\sigma(t)} - BC$. By using the Kalman rank condition (3), the system (58) becomes observable in continuous time. In this example, we investigate ϵ -observability problem for $\epsilon = 1$ and estimate initial state $x_0 = [0.2 \ 0.2]^T$ by means of a signal output sequence. In this regard, we use the 2-D sampling pattern (43) with main candidates satisfying

$$\eta = 5 \quad \text{and} \quad \xi = \frac{\pi}{16}. \quad (59)$$

Obtain that

$$\begin{aligned} P_{48} &\subset [19, 20); & P_{98} &\subset [39, 40); & P_{148} &\subset [59, 60); \\ P_{198} &\subset [79, 80); & P_{248} &\subset [99, 100); & P_{298} &\subset [119, 120]. \end{aligned}$$

Under these specifications, we consider the measured output as in Fig. 10 which is a signal measured from (58).

By using (A.6), we obtain an initial state estimation

$$\hat{x}_0 = [0.2 \ 0.2]^T,$$

which demonstrates that ϵ -observability is preserved with the proposed sampling schedule for $\epsilon = 1$. In addition, we estimate that initial state by the knowledge of the output sequence $\{y_0, y_1, \dots, y_{1500}\}$.

4.4. Acceleration driven damper

As another practical example, we consider the acceleration driven damper for suspension of vehicles as depicted in Fig. 11 (see, e.g., [36,37]). The model is as follows:

$$\begin{aligned} M\ddot{z}(t) &= -c(t)\dot{w}(t) - kw(t) + k\Delta_s - Mg \\ m\ddot{z}_t(t) &= c(t)\dot{w}(t) + kw(t) - k\Delta_s - k_t\Delta_t - mg \end{aligned} \quad (60)$$

where $w(t) = z(t) - z_t(t)$ together with $z(t)$, $z_t(t)$ and $z_r(t)$ are the vertical position of the body, the unsprung mass, and the road profile, respectively. The coefficients M and m are the quarter-car body mass and the unsprung mass (tire, wheel, brake, ...), respectively. The coefficients k and k_t are the stiffness of the suspension spring and the stiffness of the tire, respectively. The coefficients Δ_s and Δ_t are the length of the unloaded suspension spring and the tire. Finally, $c(t)$ is the damping coefficient of the passive shock absorber. As in [38], we consider a switching strategy, so that $c(t)$ can assume two values, namely $c_{\min}$ and $c_{\max}$, to be specified later on.

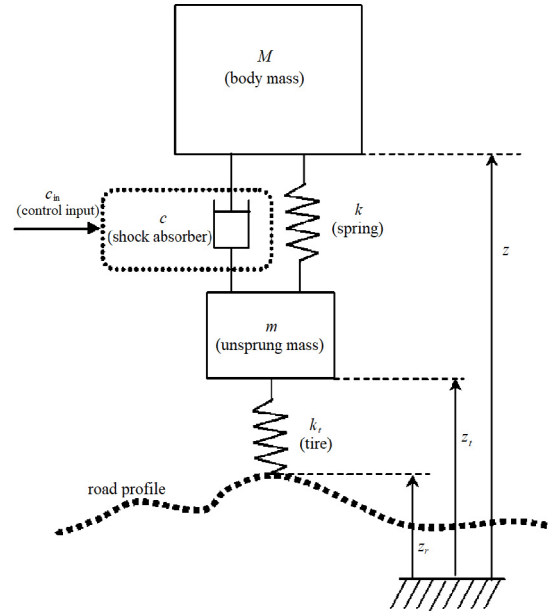


Fig. 11. A Quarter Car Model [36,37].

The variations $\delta z(t)$ and $\delta z_t(t)$ of $z(t)$ and $z_t(t)$ around an equilibrium point associated with a zero road profile yield to the linear switched system (15) with new state variables

$$\begin{aligned} x(t) &= [x_1(t) \ x_2(t) \ x_3(t) \ x_4(t)]^T \\ &= [\delta z(t) \ \delta \dot{z}(t) \ \delta z_t(t) \ \delta \dot{z}_t(t)]^T, \end{aligned}$$

and input $u(t) = z_r(t)$ which acts as a disturbance input. Apart from [38], the measurement noise is taken as $d(t) \equiv 0$ in the system (15) and the matrices are given as

$$\begin{aligned} A_1 &= \begin{bmatrix} 0 & 1 & 0 & 0 \\ -\frac{k}{M} & -\frac{c_{\min}}{M} & \frac{k}{M} & \frac{c_{\min}}{M} \\ 0 & 0 & 0 & 1 \\ \frac{k}{m} & \frac{c_{\min}}{m} & -\frac{(k+k_t)}{m} & -\frac{c_{\min}}{m} \end{bmatrix}, & B_1 &= \begin{bmatrix} 0 \\ 0 \\ 0 \\ \frac{k_t}{m} \end{bmatrix}, \\ A_2 &= \begin{bmatrix} 0 & 1 & 0 & 0 \\ -\frac{k}{M} & -\frac{c_{\max}}{M} & \frac{k}{M} & \frac{c_{\max}}{M} \\ 0 & 0 & 0 & 1 \\ \frac{k}{m} & \frac{c_{\max}}{m} & -\frac{(k+k_t)}{m} & -\frac{c_{\max}}{m} \end{bmatrix}, & B_2 &= \begin{bmatrix} 0 \\ 0 \\ 0 \\ \frac{k_t}{m} \end{bmatrix}, \\ C_1 &= \begin{bmatrix} -\frac{k}{M} & -\frac{c_{\min}}{M} & \frac{k}{M} & \frac{c_{\min}}{M} \\ 0 & 1 & 0 & -1 \end{bmatrix}, \\ C_2 &= \begin{bmatrix} -\frac{k}{M} & -\frac{c_{\max}}{M} & \frac{k}{M} & \frac{c_{\max}}{M} \\ 0 & 1 & 0 & -1 \end{bmatrix}. \end{aligned}$$

For the numerical simulations, the parameters are taken as $M = 400$ kg, $m = 50$ kg, $k = 2.0 \times 10^4$ N/m, $k_t = 2.5 \times 10^5$ N/m, $c_{\min} = 3.0 \times 10^2$ (N s)/m and $c_{\max} = 3.9 \times 10^3$ (N s)/m. To represent an arbitrary switching scenario, the switching signal is taken as

$$\sigma(t) = \begin{cases} 2, & t \in [0.3, 0.45) \cup [0.7, 0.85) \\ 1, & \text{otherwise} \end{cases} \quad (61)$$

Here, we see that matrix couples (A_1, C_1) and (A_2, C_2) satisfy the Kalman rank condition which makes each sub-system observable in continuous time. In this example, we consider ϵ -observability problem for the case $\epsilon = 0.01$ and estimate the initial state $x_0 = [0 \ 0 \ 0 \ 0]^T$ under the sampling pattern (53) with main candidates satisfying

$$\eta = 1400 \quad \text{and} \quad \xi = \frac{\pi}{4400}. \quad (62)$$

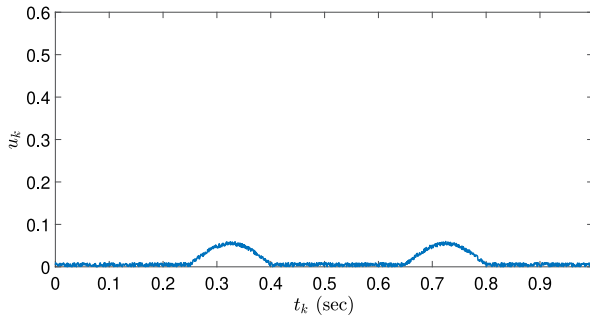


Fig. 12. Road Acceleration u_k on a Road with Two Bumps.

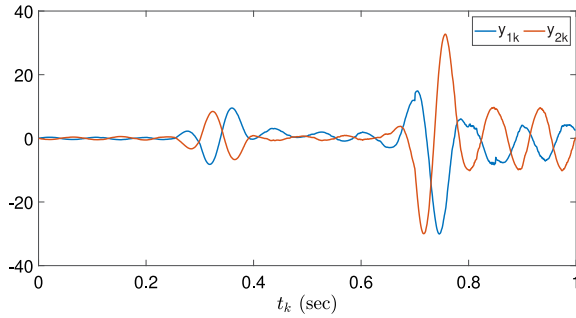


Fig. 13. Output Signal y_k Measured at each Sampling Points.

Then, we have

$$\begin{aligned} P_{68} &\subset [0.29, 0.3]; & P_{103} &\subset [0.44, 0.45]; \\ P_{162} &\subset [0.69, 0.7]; & P_{197} &\subset [0.84, 0.85]. \end{aligned} \quad (63)$$

as desired.

In order to associate a scenario of two bumps on a road, u_k is measured at each sampling time t_k for $k \in \mathbb{N}$ as it is seen in Fig. 12.

As a measured output, we presume the signal y_k is measured at each sampling point for $k \in \mathbb{N}$ as in Fig. 13.

By using (A.6), we obtain an initial state estimation

$$\hat{x}_0 = 10^{-4} \times [0.0199 \quad -0.0679 \quad 0.0008 \quad 0.1357]^T,$$

which demonstrates that ϵ -observability is preserved by the proposed sampling schedule. This estimation is achieved by means of output sequence $\{y_0, y_1, \dots, y_{1406}\}$ and input sequence $\{u_0, u_1, \dots, u_{1406}\}$.

Remark 6. In Sections 4.1 and 4.2, we take the switching controls as occurring in a periodic manner to investigate the interaction between the continuous and discrete cases of the problem. For that reason, we catch the first target at the 251th step and the second one at the 17715th step, respectively. However, from Corollary 3.6, we see that a short-cut exists by finding an appropriate switching control sequence in each problem such that the transfer could be established after a minimum number of steps satisfying (34). This means that the minimum number of steps ν which depends on the initial and final target, satisfy the inequality of $5 \leq \nu \leq 10$ and $17 \leq \nu \leq 34$ for the first and second problems respectively. In this regard, these switching controls may not be the same as periodic ones as seen in those examples.

Remark 7. In Sections 4.1 and 4.2, we realize that the stabilizing feedback gains for the continuous switched system also stabilize the switched sampled-data system. This situation may not always happen because the conditions ensuring the stability

of continuous-time and discrete-time systems are not usually the same. More precisely, continuous-time systems have to satisfy the continuous-time algebraic Riccati inequalities whereas discrete-time systems must hold the discrete-time algebraic Riccati conditions.

5. Conclusion

In this work, we study the controllability and observability problems of linear switched sampled-data systems emerged from the continuous-time invariant switched systems after the procedure of sample and zero-order hold operation by means of the non-equidistant sampling scheme. Two new notions are introduced as ϵ -controllability and ϵ -observability in the context of switched sampled-data systems. Unlike the original definitions of controllability and observability, ϵ -controllability/ ϵ -observability makes use of any initial time taken from the domain which stays ϵ apart from the switching instants in sampled-data systems. In this regard, controllability and observability properties would be achieved when ϵ approaches to zero, which requires a large number of sampling times to control/observe the system in discrete case. Because of that reason, one may prefer to use the ϵ -controllability/ ϵ -observability notion instead of the original one in the context of switched sampled-data systems to avoid such computation and the high cost of preserving the desired properties after the transfer. From the main results, ϵ -controllability and ϵ -observability properties provide a point of view to control or observe the switched sampled-data system within each of its sub-systems. In this procedure, we see that the Kalman rank condition becomes a necessary and sufficient condition for the switched sampled-data system to be ϵ -controllable/ ϵ -observable. Additionally, the suggested sampling schedule gives us the flexibility to choose proper candidates η and ξ for making its upper diameter as small as possible. More interestingly, we see in both controllability examples that the state feedback controls for the continuous-time switched systems also stabilize the sampled-data systems. This observation motivates us to pose a problem of preserving stability property of linear switched sampled data systems by the constructed sampling scheme which needs to be further demonstrated and extended in the future works. Since, the problem of controllability and observability for non-linear continuous time systems requires a more general version Kalman rank condition which varies according to the given differential system, the problem of controllability/observability becomes much more complicated for a non-linear switched system which consists of these continuous non-linear systems as its sub-systems. Hence, the sampling schedule (9) may be ineffective in the process of preserving ϵ -controllability/ ϵ -observability property for the corresponding non-linear sampled data system. This is the limitation of the presented work which needs further investigation in the future studies.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix

In this part, we shortly provide a method (similar to that used in [39]) which is performed in Sections 4.3 and 4.4 for estimating the initial state x_0 with output measurement y and input u in discrete time. In this regard, we use the proposed non-equidistant sampling pattern (9) to obtain ϵ -observability on switched linear systems. To that aim, we first define the following matrices, for each index $k \in \mathbb{N}$:

$$\begin{aligned}\tilde{B}_k &= \int_{t_k}^{t_{k+1}} e^{A_{\sigma(\tau)(t_{k+1}-\tau)} B_{\sigma(\tau)}} d\tau, \\ \tilde{A}_k &= e^{A_{\sigma(t_k)(t_{k+1}-t_k)}}, \quad \tilde{C}_k = C_k,\end{aligned}\quad (\text{A.1})$$

where $t_k \in P_m$ denotes the sampling times. Hence, the sampled-data system Σ_d becomes

$$\begin{aligned}x_{k+1} &= \tilde{A}_k x_k + \tilde{B}_k u_k, \\ y_k &= \tilde{C}_k x_k.\end{aligned}\quad (\text{A.2})$$

In this context, the ϵ -observability problem becomes a problem of estimating the initial state x_0 from the knowledge of $\{y_0, y_1, \dots, y_K\}$ and $\{u_0, u_1, \dots, u_K\}$ for some period $K \in \mathbb{N}$. Then, we have

$$\begin{bmatrix} y_0 \\ y_1 \\ \vdots \\ y_{K-1} \end{bmatrix} = \mathcal{O} x_0 + \mathcal{T} \begin{bmatrix} u_0 \\ u_1 \\ \vdots \\ u_{K-1} \end{bmatrix}\quad (\text{A.3})$$

where

$$\mathcal{O} = \begin{bmatrix} \tilde{C}_0 \\ \tilde{C}_1 \tilde{A}_0 \\ \vdots \\ \tilde{C}_K \tilde{A}_{K-1} \dots \tilde{A}_0 \end{bmatrix},\quad (\text{A.4})$$

$$\mathcal{T} = \begin{bmatrix} 0 & \dots & 0 & 0 \\ \tilde{C}_1 \tilde{B}_0 & \dots & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots \\ \tilde{C}_K \tilde{A}_{K-1} \dots \tilde{A}_1 \tilde{B}_0 & \dots & \tilde{C}_K \tilde{B}_{K-1} & 0 \end{bmatrix}.\quad (\text{A.5})$$

Here, $\mathcal{O}$ maps the initial state into the resulting output over $0, 1, \dots, K-1$ and $\mathcal{T}$ maps the input to the output over $0, 1, \dots, K-1$. Hence, if the ϵ -observability is preserved, we expect to obtain an initial state estimation of $\hat{x}_0$ for x_0 in the following manner:

$$\hat{x}_0 = \mathcal{O}^+ \left(\begin{bmatrix} y_0 \\ y_1 \\ \vdots \\ y_{K-1} \end{bmatrix} - \mathcal{T} \begin{bmatrix} u_0 \\ u_1 \\ \vdots \\ u_{K-1} \end{bmatrix} \right)\quad (\text{A.6})$$

where $\mathcal{O}^+$ is the Moore–Penrose inverse of $\mathcal{O}$ satisfying $\mathcal{O}^+ \mathcal{O} = I$ and I is the identity matrix with the appropriate dimension [34].

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